

ISE 2: Modern Full-Stack Development [MERN Authentication Project]

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GitHub Repository Link:

https://github.com/MorePrajita/Modern-Full_Stack-Development-MERN-Authentication-Project-.git

Code Description:

This is a mini-project implementing a secure User Authentication System using the MERN stack. It allows new users to register and existing users to securely log in. The project demonstrates the integration of a React frontend with a Node.js/Express backend and a MongoDB database.

Key Features

- Sign Up: Collects First Name, Last Name, Email, and Password.
- Sign In: Authenticates existing users via Email and Password.
- Security: Uses bcryptjs for password hashing (no plain-text passwords stored).
- Responsive UI: Built with Ant Design (antd) for a modern, professional look.
- Validation: Includes frontend and backend validation for required fields and email formats.

Tech Stack

- Frontend: React.js, Ant Design, Axios
- Backend: Node.js, Express.js
- Database: MongoDB (Mongoose ODM)
- Authentication: Bcrypt.js (Hashing)

CODE:

FRONTEND:

APP.js

```
import React, { useState } from 'react';

import { Form, Input, Button, Card, Typography, message, Tabs } from 'antd';

import { UserOutlined, LockOutlined, MailOutlined } from '@ant-design/icons';

import axios from 'axios';

import './App.css'; // We will add a small background tweak here


const { Title, Text } = Typography;

const App = () => {

  const [loading, setLoading] = useState(false);

  const [activeTab, setActiveTab] = useState('1');


  // Handle Form Submission

  const onFinish = async (values) => {

    setLoading(true);

    const isLogin = activeTab === '1';

    const url = isLogin ? 'http://localhost:5000/api/signin' : 'http://localhost:5000/api/signup';

    try {

      const res = await axios.post(url, values);

      message.success(res.data.message || "Operation Successful!");

    } catch (err) {

      message.error(err.response?.data?.error || "Something went wrong");

    } finally {

      setLoading(false);

    }

  };

};
```

```

return (
  <div className="auth-container">
    <Card className="auth-card" hoverable>
      <div style={{ textAlign: 'center', marginBottom: 20 }}>
        <Title level={2}>KIT's IMER</Title>
        <Text type="secondary">MCA Department Auth System</Text>
      </div>

      <Tabs
        activeKey={activeTab}
        onChange={setActiveTab}
        centered
        items={[
          { label: 'Sign In', key: '1' },
          { label: 'Sign Up', key: '2' },
        ]}
      />

      <Form name="auth_form" onFinish={onFinish} layout="vertical">
        {activeTab === '2' && (
          <>
            <Form.Item name="firstName" rules={[{ required: true, message: 'Please input First Name!'
}}}>
              <Input prefix={<UserOutlined />} placeholder="First Name" size="large" />
            </Form.Item>
            <Form.Item name="lastName" rules={[{ required: true, message: 'Please input Last Name!'
}}}>
              <Input prefix={<UserOutlined />} placeholder="Last Name" size="large" />
            </Form.Item>
          </>
        )}
      </Form>
    </Card>
  </div>
)

```

```
    <Form.Item name="email" rules={[{ required: true, type: 'email', message: 'Please input a valid Email!' }]}>
```

```
      <Input prefix={<MailOutlined />} placeholder="Email Address" size="large" />
```

```
    </Form.Item>
```

```
    <Form.Item name="password" rules={[{ required: true, message: 'Please input Password!' }]}>
```

```
      <Input.Password prefix={<LockOutlined />} placeholder="Password" size="large" />
```

```
    </Form.Item>
```

```
  <Form.Item>
```

```
    <Button type="primary" htmlType="submit" block size="large" loading={loading}>
```

```
      {activeTab === '1' ? 'Login' : 'Register'}
```

```
    </Button>
```

```
  </Form.Item>
```

```
</Form>
```

```
</Card>
```

```
</div>
```

```
);
```

```
};
```

```
export default App;
```

BACKEND:

USER.js

```
const mongoose = require('mongoose');

const UserSchema = new mongoose.Schema({
  firstName: { type: String, required: true },
  lastName: { type: String, required: true },
  email: { type: String, required: true, unique: true },
  password: { type: String, required: true }
});

module.exports = mongoose.model('User', UserSchema);
```

SERVER.js

```
const express = require('express');
const mongoose = require('mongoose');
const cors = require('cors');
const bcrypt = require('bcryptjs');
const User = require('./models/User');

const app = express();
app.use(express.json());
app.use(cors());

// Connect to MongoDB (Replace with your local or Atlas URI)
mongoose.connect('mongodb://127.0.0.1:27017/mern_db')
  .then(() => console.log("MongoDB Connected"))
  .catch(err => console.log(err));

// SIGN UP ROUTE
app.post('/api/signup', async (req, res) => {
```

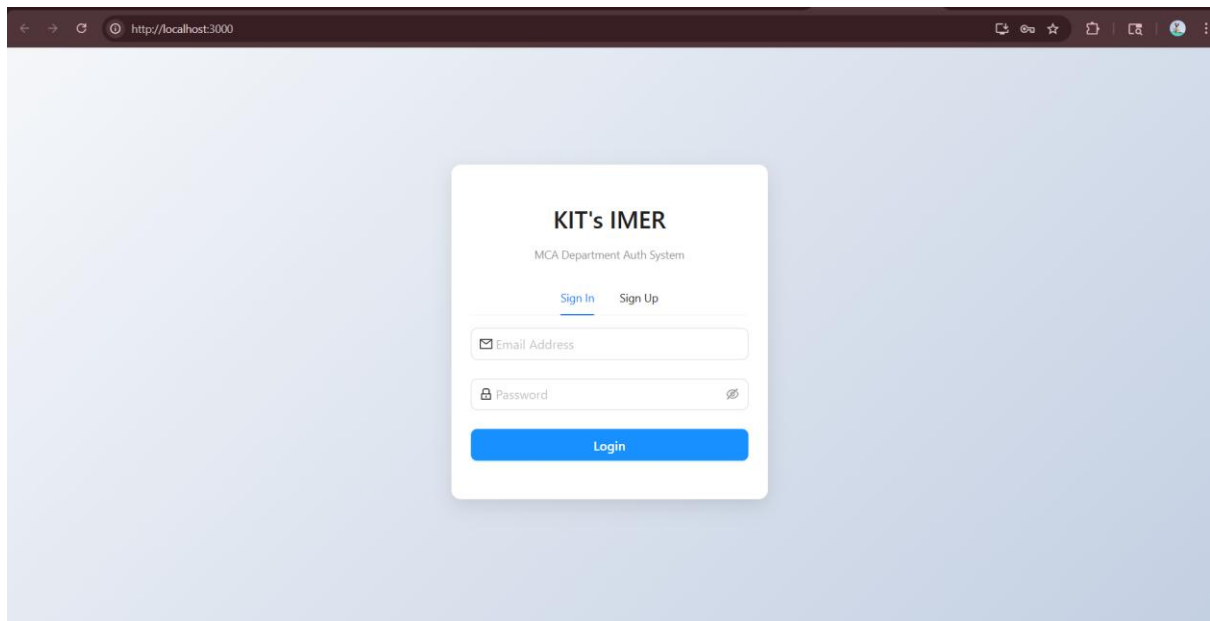
```
try {  
  const { firstName, lastName, email, password } = req.body;  
  const hashedPassword = await bcrypt.hash(password, 10);  
  const newUser = new User({ firstName, lastName, email, password: hashedPassword });  
  await newUser.save();  
  res.status(201).json({ message: "User Created Successfully" });  
} catch (err) {  
  res.status(400).json({ error: "Email already exists or invalid data" });  
}  
});
```

// SIGN IN ROUTE

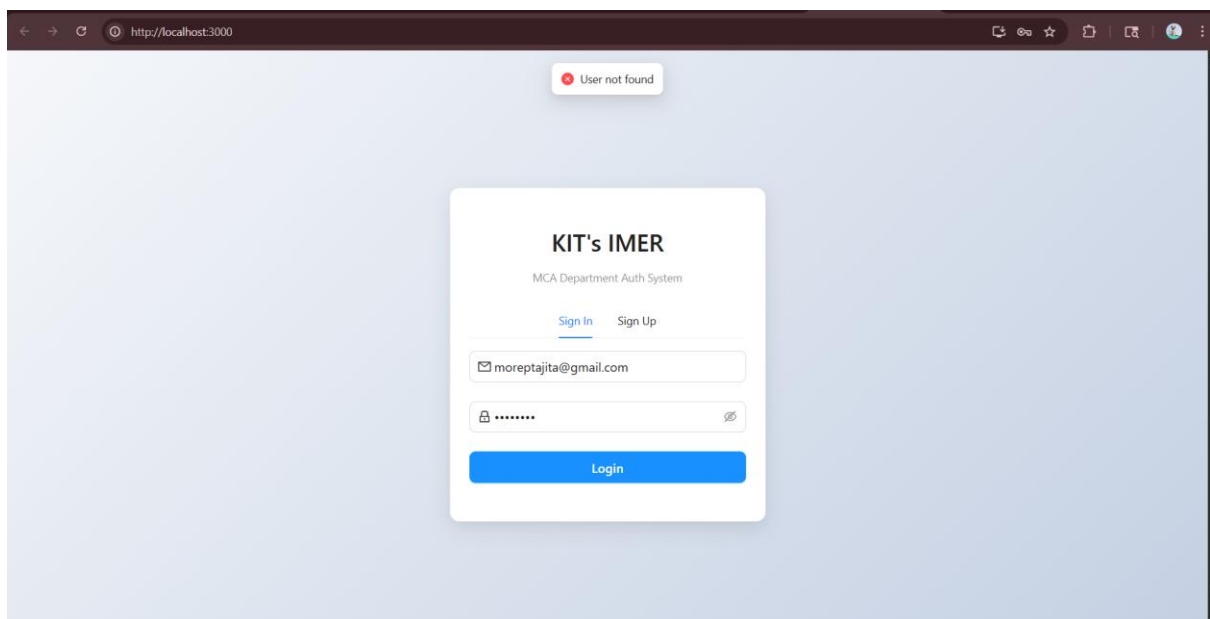
```
app.post('/api/signin', async (req, res) => {  
  const { email, password } = req.body;  
  const user = await User.findOne({ email });  
  if (!user) return res.status(400).json({ error: "User not found" });  
  
  const isMatch = await bcrypt.compare(password, user.password);  
  if (!isMatch) return res.status(400).json({ error: "Invalid credentials" });  
  
  res.json({ message: "Login Successful", user: user.firstName });  
});
```

```
app.listen(5000, () => console.log("Server running on port 5000"));
```

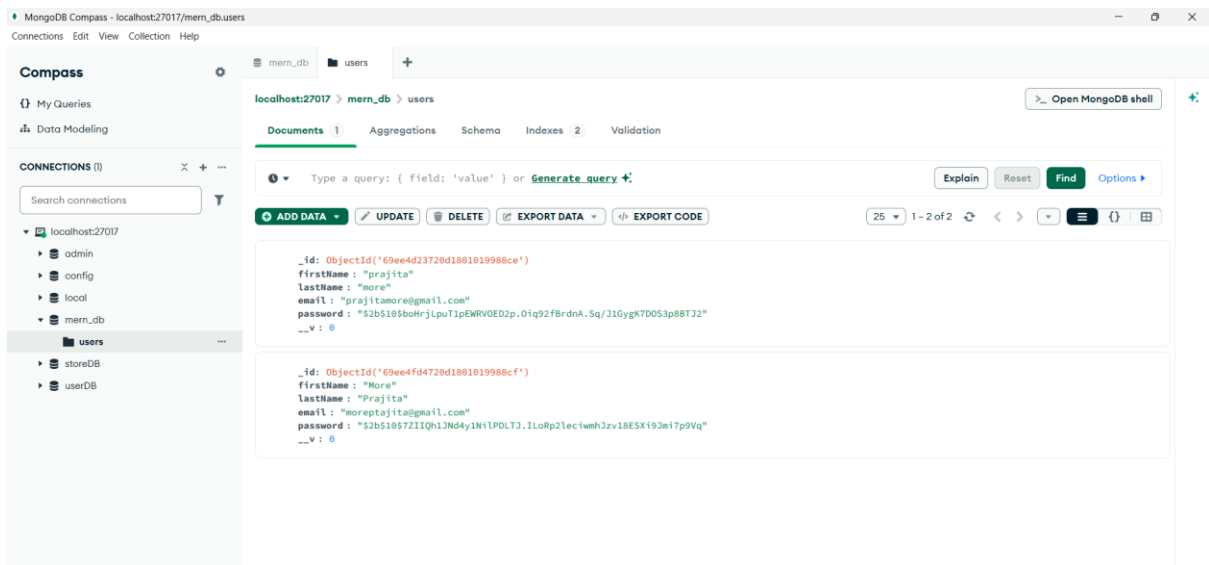
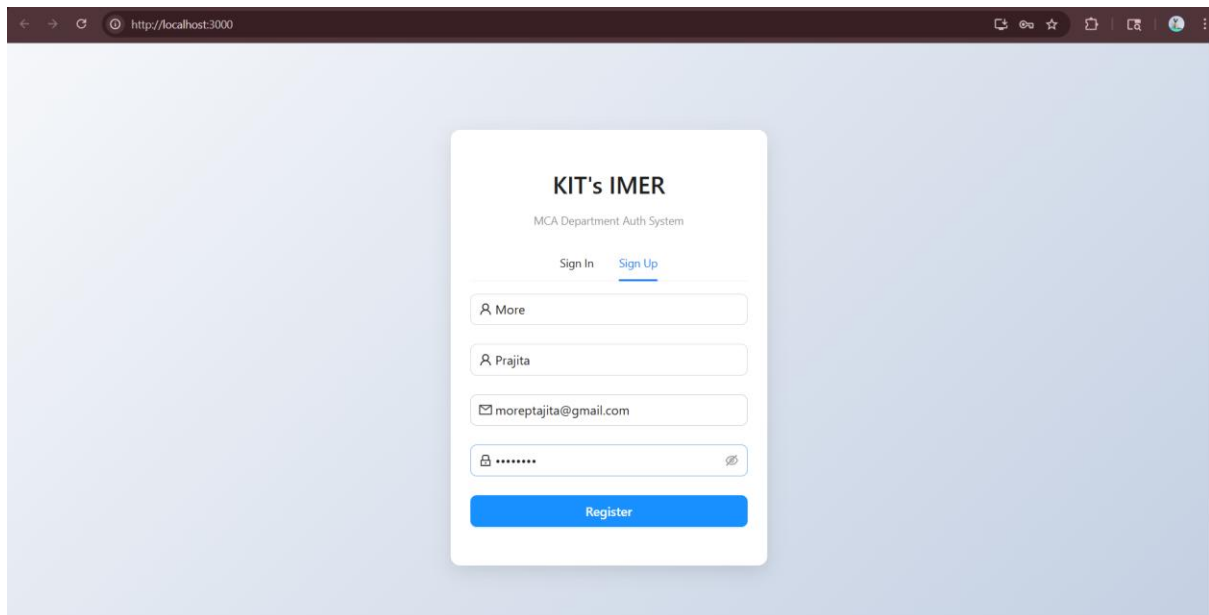
OUTPUT:



A screenshot of a web browser displaying the login page for 'KIT's IMER'. The browser's address bar shows 'http://localhost:3000'. The page has a light blue gradient background. In the center, there is a white card with the title 'KIT's IMER' and the subtitle 'MCA Department Auth System'. Below the title are two links: 'Sign In' (underlined) and 'Sign Up'. There are two input fields: 'Email Address' with an envelope icon and 'Password' with a lock icon and a toggle for visibility. A blue 'Login' button is at the bottom of the card.



A screenshot of the same web browser showing the login page after an attempt. A red error message 'User not found' is displayed at the top. The login card is identical to the first screenshot, but the 'Email Address' field now contains 'moreptajita@gmail.com' and the 'Password' field contains seven dots. The 'Login' button remains blue.



```
• PS E:\MCA\SEM2\FullStack\ISE2\client> cd..
• PS E:\MCA\SEM2\FullStack\ISE2> git init
Initialized empty Git repository in E:/MCA/SEM2/FullStack/ISE2/.git/
• PS E:\MCA\SEM2\FullStack\ISE2> git add .
warning: adding embedded git repository: client
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint:   git submodule add <url> client
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint:   git rm --cached client
hint:
hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"
warning: in the working copy of 'server/package-lock.json', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'server/package.json', LF will be replaced by CRLF the n
ext time Git touches it
• PS E:\MCA\SEM2\FullStack\ISE2> git commit -m "Complete MERN Authentication Project"
[master (root-commit) a5b0b33] Complete MERN Authentication Project
 8 files changed, 1292 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 New Microsoft Word Document.docx
 create mode 160000 client
 create mode 100644 server/models/User.js
 create mode 100644 server/package-lock.json
 create mode 100644 server/package.json
 create mode 100644 server/server.js
 create mode 100644 ~$w Microsoft Word Document.docx
• PS E:\MCA\SEM2\FullStack\ISE2> git remote add origin https://github.com/MorePrajita/Mode
rn-Full_Stack-Development-MERN-Authentication-Project-.git
• PS E:\MCA\SEM2\FullStack\ISE2> git branch -M main
• PS E:\MCA\SEM2\FullStack\ISE2> git push -u origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 12 threads
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS E:\MCA\SEM2\FullStack\ISE2> git add .

hint: git submodule add <url> client

hint:

hint: If you added this path by mistake, you can remove it from the

hint: index with:

hint:

hint: git rm --cached client

hint:

hint: See "git help submodule" for more information.

hint: Disable this message with "git config set advice.addEmbeddedRepo false"

warning: in the working copy of 'server/package-lock.json', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'server/package.json', LF will be replaced by CRLF the next time Git touches it

● PS E:\MCA\SEM2\FullStack\ISE2> git commit -m "Complete MERN Authentication Project"

[master (root-commit) a5b0b33] Complete MERN Authentication Project

8 files changed, 1292 insertions(+)

create mode 100644 .gitignore

create mode 100644 New Microsoft Word Document.docx

create mode 160000 client

create mode 100644 server/models/User.js

create mode 100644 server/package-lock.json

create mode 100644 server/package.json

create mode 100644 server/server.js

create mode 100644 ~\$w Microsoft Word Document.docx

● PS E:\MCA\SEM2\FullStack\ISE2> git remote add origin https://github.com/MorePrajita/Modern-Full_Stack-Development-MERN-Authentication-Project-.git

● PS E:\MCA\SEM2\FullStack\ISE2> git branch -M main

● PS E:\MCA\SEM2\FullStack\ISE2> git push -u origin main

Enumerating objects: 11, done.

Counting objects: 100% (11/11), done.

Delta compression using up to 12 threads

Compressing objects: 100% (9/9), done.

Writing objects: 100% (11/11), 12.81 KiB | 820.00 KiB/s, done.

Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

To https://github.com/MorePrajita/Modern-Full_Stack-Development-MERN-Authentication-Project-.git

* [new branch] main -> main

branch 'main' set up to track 'origin/main'.

○ PS E:\MCA\SEM2\FullStack\ISE2> □